

Which Design Supports Cause?

AP Statistics · Unit 1 · Topic 1.13 · TEACHER KEY

CED TETHER

- **Skill 1.C** — Describe an appropriate method for gathering and representing data.
- **Skill 1.B** — Identify key and relevant information to answer a question or solve a problem.
- **LO VAR-3.A** — Identify the components of an experiment. [Skill 1.C]
- **EK VAR-3.A.1** — The experimental units are the individuals (which may be people or other objects of study) that are assigned treatments.
- **EK VAR-3.A.2** — An explanatory variable (or factor) in an experiment is a variable whose levels are manipulated intentionally. The levels or combination of levels are called treatments.
- **EK VAR-3.A.3** — A response variable in an experiment is an outcome from the experimental units that is measured after the treatments have been administered.
- **LO VAR-3.B** — Describe elements of a well-designed experiment. [Skill 1.B]
- **EK VAR-3.B.1** — A well-designed experiment includes comparison, random assignment, replication, and control of potential confounding variables where appropriate.
- **EK VAR-3.C.1** — Random assignment tends to balance uncontrolled variables so that differences in responses can be attributed to treatments.
- **EK VAR-3.C.4** — In a double-blind experiment neither the subjects nor the research team members who interact with them know which treatment a subject is receiving.
- **EK VAR-3.C.5** — A control group receives no treatment of interest or an inactive treatment so the treatment of interest can be evaluated.

Essential question: Which design features let a researcher attribute a response difference to a treatment?

DOK-3 skill: 4.A · **FRQ pattern:** experiment-principles-which-design-supports-cause

DOK rationale: Part (c) compares two designs against a causal claim, links random assignment, control, and blinding to specific threats, explains a concrete alternative cause, and bounds generalization from volunteers.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- What exactly did the researcher manipulate, and what was measured afterward?
- Which feature addresses preexisting differences between early and late volunteers?
- What does the control group reveal that one treatment group cannot?
- Who was randomized, and from what larger population were they not randomly selected?

[WARN] LOOK FOR

- Treating signup order or equal group size as random assignment.
- Calling the study representative of teenagers because its treatments were randomized.
- Naming blinding without saying whose expectations or measurements it protects.

SCORING PART (c) — E / P / I

Expected elements

- **design-and-assignment** (required) — Chooses Design 2 and links random assignment to balancing other factors
- **signup-alternative** (required) — Links a plausible early/late volunteer difference to reaction time
- **volunteer-limit** (required) — Explains why volunteers cannot automatically represent all teenagers

E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

Common mistakes

- Calling Design 1 randomized because students arrived in an unpredictable order
- Treating equal group sizes as a substitute for random assignment
- Saying identical cups alone blind participants without the matched drink and concealed assignment
- Confusing random assignment among volunteers with random selection of teenagers

[STAR] THE PROBLEM — annotated

Suppose a researcher recruits 30 student volunteers to study whether an energy drink changes reaction time.

Design 1: The first 15 students to sign up receive the energy drink; the last 15 receive water. A tester who knows each drink measures reaction time in milliseconds.

Design 2: A coin flip assigns each volunteer to the energy drink or water. The water is flavored to match, both drinks are served in identical opaque cups, and neither the volunteer nor the tester knows which drink was assigned.

Two energy-drink designs

#	Drink allocation	Tester told drink?
1	First/last signup	Yes
2	Coin flip	No

FIRST TAKE — one sentence before you work the parts

Which design would make the comparison between drinks fairer? Give one reason.

Key: Not graded. Expect students to notice the coin flip before they can explain why it matters.

[BOOK] PRINCIPLES OF EXPERIMENTAL DESIGN (keep on the page)

A well-designed experiment uses **comparison**, **random assignment**, **replication**, and control of potential confounding variables. A **control group** supplies a baseline. Random assignment tends to balance uncontrolled variables so a difference can be attributed to treatment. **Blinding** reduces changes caused by expectations or by how responses are measured. Selection still limits generalization.

DOK 1 (a) Name the experimental units and the response variable, with units.

Key: The 30 volunteers are the experimental units. The response is reaction time in milliseconds.

DOK 2 (b) Compare how treatments are assigned in the two designs. Explain how one other feature of Design 2 helps the comparison.

Key: Design 1 uses signup order; Design 2 uses random assignment by coin flip. Accept either additional feature with its job: matched water supplies a comparison while making the drinks hard to distinguish, or concealed assignments reduce expectation and tester effects.

DOK 3 (c) Which design gives stronger evidence about whether the drink causes a change? Explain one other reason the groups in Design 1 might react differently. State why the result would not automatically describe all teenagers.

Key: Design 2 gives stronger causal evidence because random assignment tends to balance other factors. In Design 1, earlier volunteers might differ in sleep, motivation, or baseline reaction time, which could also explain the observed difference. Neither group of volunteers automatically represents all teenagers; treatment assignment is not random selection of participants.